KAUSTUBH WATANE

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Portfolio | Linkedin | GitHub | Substack

SUMMARY

AI Engineer at GE HealthCare building production-grade Generative AI systems for engineering and regulated environments. Worked across enterprise RAG, AI-assisted verification, and engineering document automation, with a focus on reliability, evaluation, and secure deployment.

WORK EXPERIENCE

AI Engineer (Trainee), GE HealthCare, Bangalore **Aug 2025 - Present**

Nebula AI — Enterprise AI Workspace

- Designed and deployed a secure, on-premise RAG platform for GE HealthCare engineering teams, adopted by 50%+ of the team and contributing to 30%+ workflow efficiency gains.
- Built hybrid retrieval using BM25, BGE-M3 embeddings, Reciprocal Rank Fusion, and cross-encoder reranking to deliver accurate, source-grounded answers from engineering documents.
- Extended the platform with multimodal document understanding, real-time voice interaction, AI-assisted coding, smart notes, and access-controlled administration.

TestMatrix AI — V&V Test Automation

- Developing an AI-assisted V&V test-case generation system for medical-device engineering aligned with IEC 62304 and FDA 21 CFR 820.
- Designed layered evaluation using hallucination detection, LLM verification, quality gates, guardrails, and multi-metric evaluation to improve reliability before human review.

Atlas AI — Engineering Document Generator

- Developing an offline AI platform that generates structured DRS, SSRS, and SWRS engineering specifications from short product briefs using multi-stage LangGraph workflows.
- Designed parallel drafting, automated UR-ID allocation, compliance validation, hallucination checks, and requirements traceability, reducing specification development from weeks to minutes.

TECHNICAL SKILLS

- **Generative AI / LLMs:** Generative AI, LLMs, RAG, Agentic AI, AI Agents, Multi-Agent Systems, LangGraph, LangChain, Prompt Engineering, Tool Calling, LLM Evaluation, Guardrails, Multimodal AI
- **AI Retrieval / NLP:** Hybrid Search, Semantic Search, BM25, Dense Retrieval, Embeddings, Reranking, Vector Databases, Qdrant, Document Parsing, OCR
- **AI Engineering / Backend:** Python, FastAPI, Async Programming, REST APIs, Pydantic, SQLAlchemy, AI System Architecture, WebSockets
- **Voice AI:** Speech-to-Text (STT), Text-to-Speech (TTS), Voice Activity Detection, Real-Time Streaming
- **Data / Databases:** PostgreSQL, Redis, SQL, Firebase
- **Cloud & Deployment:** Docker, Azure AI, AWS, Linux, On-Premise AI Deployment, Ollama, Nginx
- **Frontend / Systems:** React, React Native, TypeScript, JavaScript, C, Embedded C

PROJECTS

Luna AI — AI Companion Platform 🔗

- Built and deployed a cross-platform AI companion app using React Native and Expo, with a FastAPI + PostgreSQL backend, integrating Gemini, Claude, Grok, and OpenAI APIs through a unified conversational interface.
- Implemented AI chat, voice interactions, journaling, games, and personalized companion features; conversation state and user data persisted via PostgreSQL.
- Independently architected the full stack — React Native frontend, FastAPI backend, API integration layer, and Play Store deployment pipeline — achieving 500+ users post-launch.

Netra AI — Quantitative Trading Signal & Market Intelligence Platform 🔗

- Engineered a full-stack algorithmic trading intelligence system for Indian equity markets (NSE/NIFTY 50) combining classical ML, technical indicators, and automated execution pipelines.
- Developed a multi-factor composite signal engine leveraging an Optuna-tuned XGBoost classifier (time-series cross-validation) blended with Supertrend, RSI, MACD, and volume confirmation to generate BUY/SELL/HOLD signals.
- Built a quantitative backtesting engine modeling real-world Indian market transaction costs (STT, brokerage, GST, stamp duty) with Sharpe ratio, max drawdown, and win-rate analytics.
- Designed a Bloomberg Terminal-style dark UI using Next.js 15, TypeScript, TailwindCSS, and TradingView Lightweight Charts for real-time candlestick rendering, option chain PCR tracking, and market breadth visualization.

CERTIFICATIONS

- **Building AI Products: Architecture and Orchestration Professional Certificate** — AI system architecture, agent orchestration, LLM application design
- **Microsoft Azure AI Essentials Professional Certificate —** Azure AI services, cloud AI/ML deployment, responsible AI
- **Docker Foundations Professional Certificate —** containerization, Docker Compose, image optimization

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EDUCATION

PG Diploma In Embedded System and Design — **Aug 2024 - March 2025**

CDAC ACTS, Pune — Grade A

Bachelor of Electronics and Telecommunication Engineering — **June 2019 - June 2023**

Government College of Engineering, Karad — CGPA 7.83/10
